Ferrofluid under Field and Flow

Alignment (A), Steady Morphology (B), Oscillatory Morphology (C),
Dynamic Shape Response (D), and Extended Forcing (E)
Phase-field MRT–Lattice-Boltzmann study

Project results synthesis

8 October 2026

Executive result. One delayed microscopic process—Brownian rotation of magnetic nanoparticles—organises every result in this project. In a uniform ferrofluid it changes the resistance to flow (A). At an interface it competes with surface tension and changes droplet shape (B). Under an alternating field the same response delay weakens and delays the shape forcing (C–D). Under more complicated forcing it provides useful control knobs, but rotating-field conclusions remain preliminary (E).

1. The physical picture shared by A–E

Think of the ferrofluid as liquid carrying tiny compass needles. A field $\mathbf{H}$ tries to align their collective magnetisation $\mathbf{M}$. The particle response is not instantaneous: it relaxes over the Brownian time τ_B. The field-strength parameter is

$$\xi = \frac{\mu_0 m H}{k_B T}, \qquad \mathbf{M} = \phi M_s L(\xi)\,\frac{\mathbf{H}}{H}, \qquad L(\xi) = \coth\xi - \frac{1}{\xi}.$$

The governing competition for a drop is, schematically,

$$\underbrace{\mu_0 M^2}_{\text{magnetic traction}} \quad \text{vs.} \quad \underbrace{\sigma/R}_{\text{surface tension}} \quad \text{and} \quad \underbrace{\eta\dot{\gamma}}_{\text{flow-induced stress}}.$$

The dimensionless drive frequency $\omega\tau_B$ says whether the particle magnetisation can follow the field: $\omega\tau_B \ll 1$ is near quasi-static; $\omega\tau_B \gtrsim 1$ gives an appreciable phase lag. Shape is measured with the Taylor deformation

$$D = \frac{a-b}{a+b},$$

where a and b are the principal semi-axes obtained from the phase-field second moments. Thus $D = 0$ is circular and larger positive D means a more elongated droplet.

Numerical model. The calculations use a 2D MRT lattice-Boltzmann fluid solver coupled to a diffuse-interface phase field, a magnetostatic solve, and a finite-response magnetisation equation. Standard droplet runs use a 50×50 domain, radius $R = 10$, and diffuse-interface width about four cells; rotating-field tests also use a 100×100 domain with $R = 15$. The ϕ-mass is conserved to printed precision in healthy morphology runs, and the magnetostatic residual is approximately 10^{-6}.

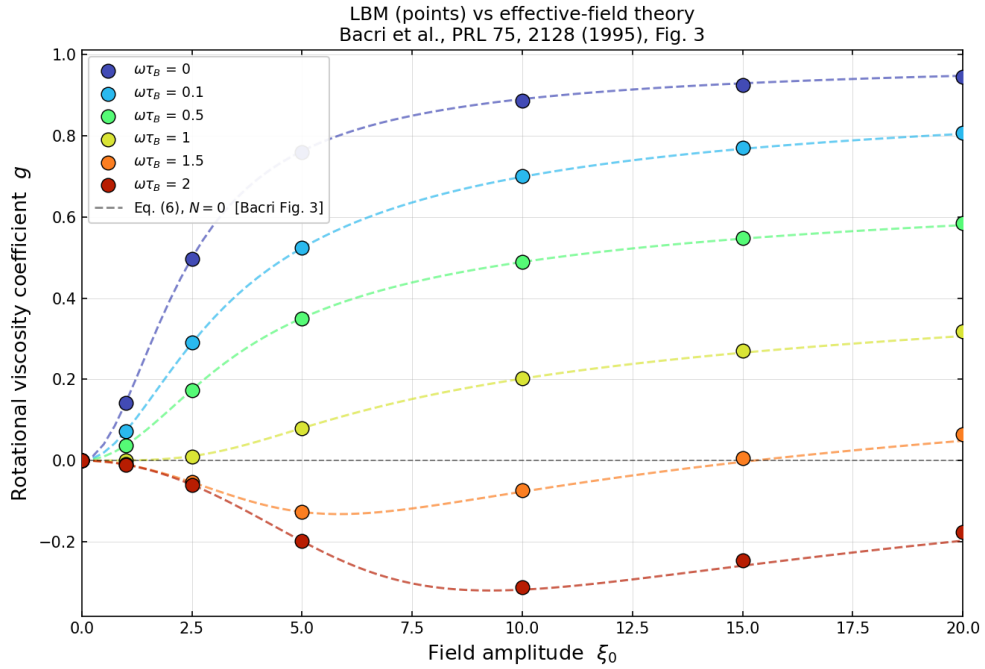
2. Section A — alignment and rotational viscosity

Question. Does the dynamic magnetisation model reproduce the known magnetoviscous response of a filled ferrofluid in pressure-driven flow?

The zero-field numerical viscosity is $\eta_0 = 0.0500386$, only 0.077% above the specified 0.0500000. Across 42 field/frequency cases, the measured dimensionless magnetoviscous contribution g agrees closely with the appropriate demagnetisation-corrected MRSh theory: maximum absolute discrepancy 0.0068 and RMS discrepancy 0.0029.

At $\xi_0 = 5$, increasing $\omega\tau_B$ changes g from +0.759 (DC) through +0.350 (0.5), +0.079 (1.0), −0.126 (1.5), and −0.197 (2.0). The field-induced contribution can therefore lower the effective flow resistance at sufficiently rapid forcing. This is *not* negative total viscosity: the total viscosity stays positive.

At $\xi_0 = 5$, $\omega\tau_B = 1.5$, the domain-averaged magnetisation lags the field by 53.5°, sensibly close to $\arctan(1.5) = 56.3°$. The weak imposed shear produces only a 0.0163° variation of alignment angle across the channel, so the observed phase lag is genuinely a temporal relaxation effect, not a wall-shear artifact.



Interpretation note. Section A is the validation foundation for the droplet work: the microscopic lag used in B–E is quantitatively behaving as intended.

3. Section B — steady-field droplet morphology

B1: field strength. With no imposed flow, the droplet remains field-aligned and deformation rises monotonically with material saturation magnetisation:

M_s	.005	.010	.015	.020	.025	.030	.035	.040
D	.0025	.0228	.0628	.1029	.1508	.2107	.2939	.3492

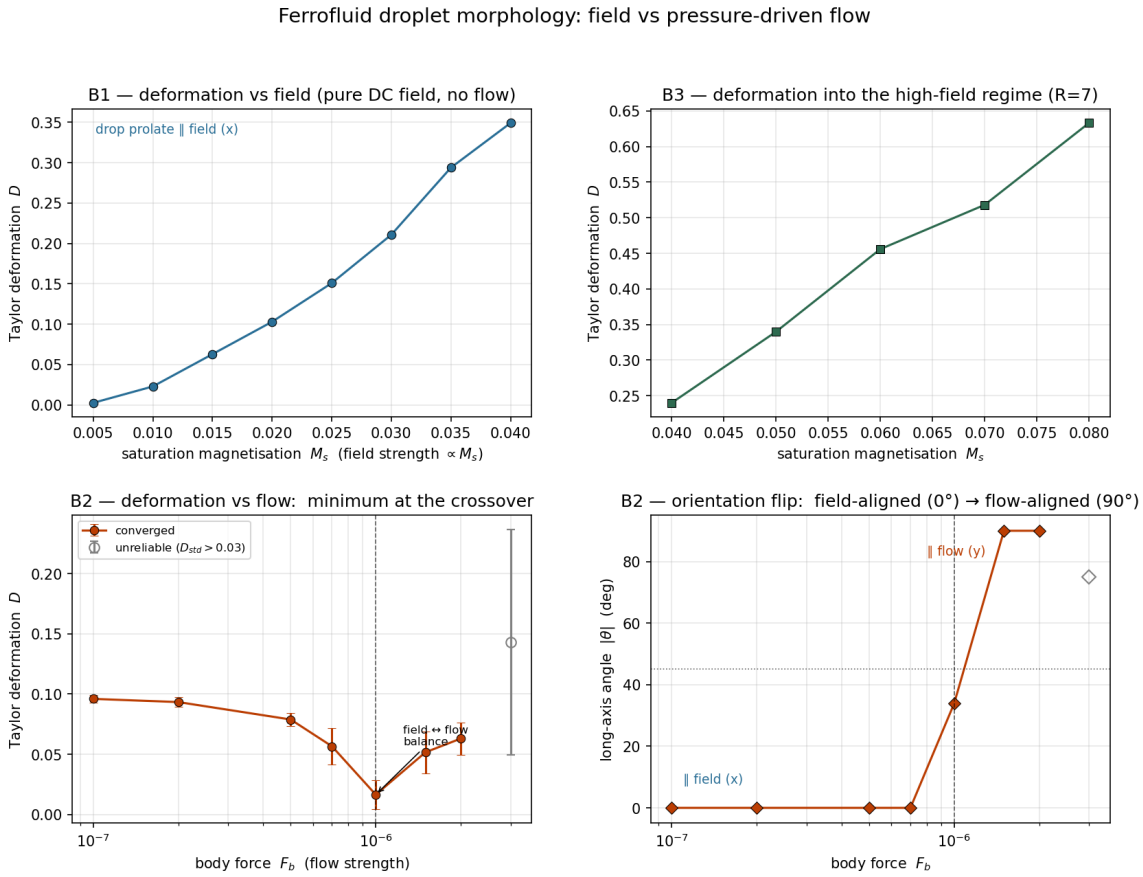
This is the expected magnetic-traction versus surface-tension result: stronger magnetisation produces a more prolate drop.

B2: field–flow crossover. At fixed $M_s = 0.02$, increasing the x-body-force parameter F_b reduces the deformation to a near-round state and then changes the principal-axis orientation:

F_b	10^{-7}	2×10^{-7}	5×10^{-7}	10^{-6}	2×10^{-6}
$\overline{D}$	.0960	.0934	.0789	.0165	.0631
$\lvert\theta\rvert$	0°	0°	0°	33.9°	90°

The robust empirical observation is a shape-balance point near $F_b \simeq 10^{-6}$. The angle is naturally noisy there because a nearly circular object has no well-defined long axis.

B3: high field. For the smaller $R = 7$ drop, D increases from .240 at $M_s = .04$ to .633 at $M_s = .08$, with no pinch-off. This does *not* determine a breakup threshold; it says only that breakup lies beyond the scanned range.



Interpretation note. The B trends are credible qualitatively. The exact crossover and the physical interpretation of the $0° \to 90°$ flip need a resolution/confinement study before being presented as precision physics.

4. Section C — oscillating-field morphology

Question. Does a sinusoidal field with the same peak amplitude as a DC field produce the same mean shape? No.

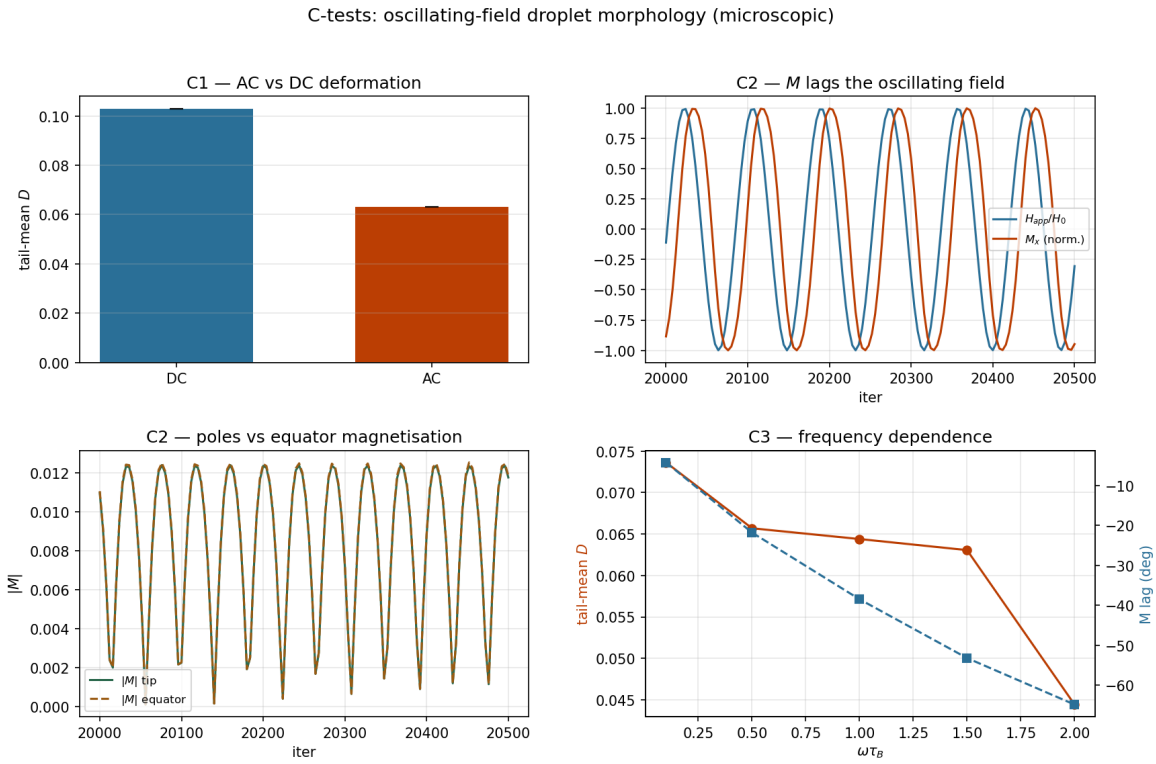
For matched conditions ($M_s = .02$, $\xi_0 = 5$, $\omega\tau_B = 1.5$), the DC case gives $\overline{D} = 0.10292$, while AC gives $\overline{D} = 0.06306$. Therefore

$$\frac{\overline{D}_{\mathrm{AC}}}{\overline{D}_{\mathrm{DC}}} = 0.613,$$

or about 39% less mean deformation under AC. The plain-language reason is that magnetic shape forcing is approximately quadratic in the magnetisation. A reversing field spends part of each cycle being followed imperfectly by $\mathbf{M}$, so its time-averaged stretching traction is smaller than the traction of a steady equal-peak field.

The frequency sweep makes the connection explicit:

$\omega\tau_B$	.1	.5	1.0	1.5	2.0
$\overline{D}$	.0736	.0657	.0644	.0631	.0444
|M lag|	4.2°	21.8°	38.5°	53.2°	64.8°



5. Section D — dynamic shape response within an AC cycle

Section C establishes that AC changes the mean shape. Section D asks the more microscopic question: *how does the shape move during a cycle?*

The dense centre run ($\xi_0 = 5$, $\omega\tau_B = 1.5$) shows a mean $\overline{D} = 0.0630595$ with only 1.06×10^{-5} half-range oscillation in D. In contrast, mean interfacial magnetic stress is 1.61×10^{-5} and ranges from essentially zero to 2.94×10^{-5}. Thus the force oscillates strongly, but the interface responds only weakly within a single period: it is a mechanical low-pass filter.

The deformation spectrum has a dominant 2ω peak (6.22×10^{-3} in the reported FFT units), much larger than the ω peak (3.89×10^{-5}). That is the expected signature of a response driven by H^2 or M^2: reversing the field reverses the moment but not the stretching magnitude. The fitted deformation phase relative to H^2 is about 74° in magnitude, showing a substantial interfacial-response delay.

Interpretation note. A useful presentation sentence is: "magnetisation lags the field, and the droplet shape lags the magnetic stress still further." Do not instead claim a permanent pole-versus-equator magnetisation enhancement: the tail mean $M_{\text{tip}}/M_{\text{eq}} = 0.9918$ is slightly below one, although it varies over a cycle.

6. Section E — extended forcing

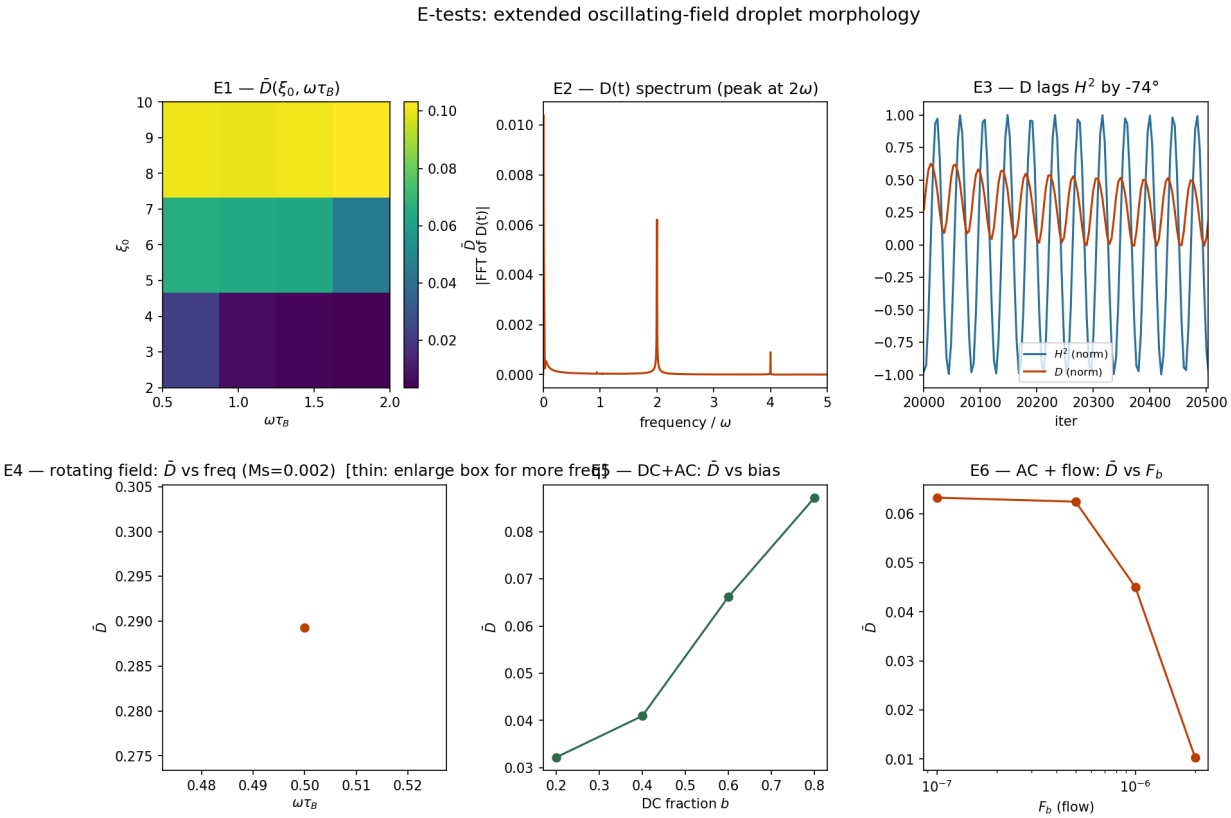
E1, amplitude–frequency map. At low amplitude $\xi_0 = 2$, $\overline{D}$ falls strongly with frequency: .022, .007, .005, .004 at $\omega\tau_B = .5, 1, 1.5, 2$. At intermediate amplitude $\xi_0 = 5$, it is .066, .064, .063, .044. At $\xi_0 = 10$, it is approximately .101, .100, .101, .103 over the same sampled frequencies. The model therefore predicts a strong frequency penalty at weak/intermediate drive, while the high-amplitude cases are nearly frequency-independent over this particular grid.

E5, DC-biased AC. Adding a DC component restores mean stretching: at DC fractions .2, .4, .6, .8, the mean deformations are .0322, .0409, .0662, .0872. Bias provides a direct control knob for mean shape.

E6, AC plus flow. Weak body force barely changes the AC shape: $\overline{D} = .0633, .0625$ at $F_b = 10^{-7}, 5\times10^{-7}$. Larger forcing reduces it to .0451 at 10^{-6} and .0104 at 2×10^{-6}; the latter also develops a mean angle of

28.5°.

E4, rotating field—preliminary only. The present rotating-field data cannot yet support a frequency curve. Ten runs diverged and nine were rejected for boundary contact. Only one clean point remains: $M_s = .002$, $\omega\tau_B = .5$, $\overline{D} = .2893$, mean $|\theta| = 25.6°$. Even available large-box runs show wall contact or divergence. This is a numerical/domain-size limitation, not evidence for a completed physical trend.



7. What can be claimed now, and what should come next

Supported conclusions.

1. The dynamic magnetisation model reproduces the targeted magnetoviscous response and the expected phase lag.
2. A DC field elongates the ferrofluid droplet monotonically over the tested field range.
3. AC fields produce substantially less mean deformation than matched DC fields, and the reduction tracks increasing magnetisation lag.
4. The instantaneous deformation response is predominantly at 2ω and delayed relative to magnetic stress.
5. DC bias provides a clear way to tune the AC mean deformation.

Priority next calculations.

1. Repeat the B2 neighbourhood around $F_b \approx 10^{-6}$ at finer resolution and in a larger domain; report convergence of D and angle.
2. Extend B3 in a larger box to identify a true pinch-off threshold rather than merely a lower bound.
3. Rebuild E4 with a larger domain, reduced stable field strength, centroid tracking, and automatic exclusion of every boundary-contact case.

4. Recheck the local tip/equator diagnostic before using it as a physical mechanism in a report or presentation.

Data basis: 42 bulk Poiseuille cases, 70 morphology summary files, their solver logs, and the A–C project documentation available in the project folder. Numerical values are tail means unless stated otherwise.